

Why We Need Retrieval-Augmented Generation?

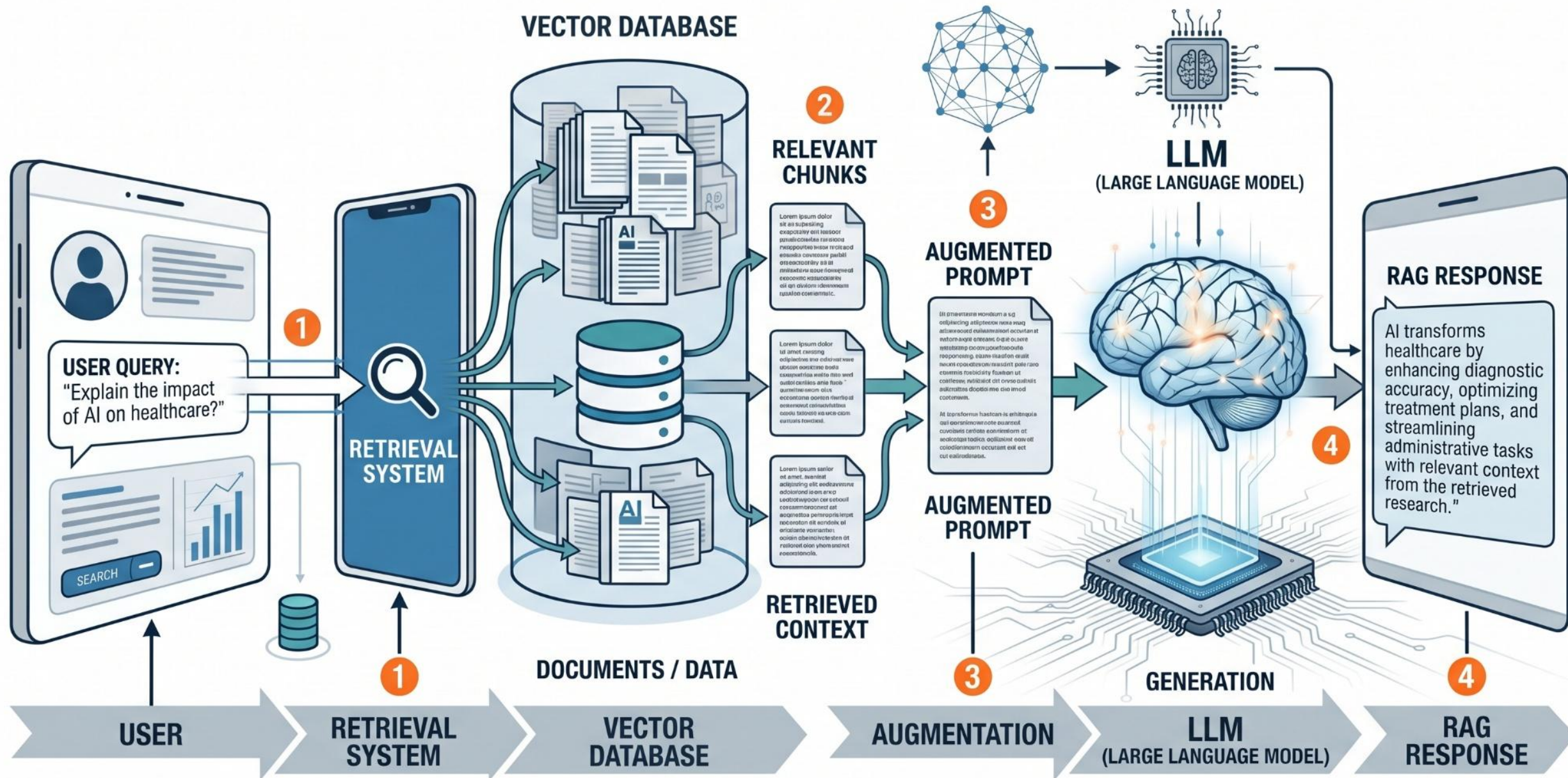
Problems

- **Knowledge cutoff:** Models are frozen at their training cutoff date.
- **No Private data:** An LLM does not know your company's internal PDFs, policies, or personal notes.
- **Hallucinations:** When unsure, LLMs can confidently make things up.

The Solution: RAG

Instead of forcing the LLM to memorize everything, give it an "Open-Book Exam."

We fetch relevant private documents first, then hand them to the LLM to generate an answer based only on facts.



What is RAG

- **Concept:** RAG is an AI architecture that connects Large Language Models (LLMs) to external knowledge sources (PDFs, databases, web pages).”
- **Step-by-Step Breakdown**
 - **Retrieve:** The user asks a question. The system searches a Vector Database for the most relevant document passages.
 - **Augment :** The system combines the user's question and the retrieved facts into a single master prompt.
 - **Generate:** The LLM reads the combined prompt and generates a accurate response using only the provided facts.

What is Word Embedding?

- **Concept:** An Embedding turns text (words, sentences, or paragraphs) into a list of numbers (a vector). Words with similar meanings end up close together in mathematical space.
- **Example**
 - **Text:** Puppy vs Dog vs Car
 - **Embedding Space:** "Puppy" and "Dog" sit right next to each other. "Car" sits far away.
- **Types of Embedding Spaces**
 - **Static Embedding Spaces:** The word "bank" has the exact same point in space Such as **Word2Vec** and **GloVe**.
 - **Contextual Embedding:** Coordinates shift based on surrounding context such as **BERT**.
 - **Sentence Embedding:** Maps entire paragraphs into a single vector representation.

What is Vector Database?

- **Concept:** A Vector Database is a specialized storage engine designed to store, index, and query high-dimensional vector embeddings (lists of numbers representing semantic meaning).
- Vector databases use mathematical formulas to measure how "close" two vectors are to each other in vector space.
- Calculating exact distances against millions of vectors takes too long. Indexing speeds this up using Approximate Nearest Neighbor (ANN) search.
- **Examples:** ChromaDB, Pinecone, pgvector (PostgreSQL), Qdrant, Milvus and FAISS

Vector Databases

Vector Database	Pricing (Free vs. Paid)	Local vs. Cloud
ChromaDB	Free (Fully free, open-source)	Local (Runs in-process/embedded or as a local server)
FAISS	Free (Fully free library by Meta)	Local (Runs entirely on local CPU/GPU)
Pinecone	Paid (Has a limited free tier, then usage-based SaaS)	Cloud-Based (Managed SaaS only, no local hosting)
Qdrant	Hybrid (Free if self-hosted; Paid for Qdrant Cloud)	Hybrid (Can run locally in Docker/server OR managed cloud)
Milvus	Hybrid (Free if self-hosted; Paid as Zilliz Cloud)	Hybrid (Can run locally via Docker/K8s OR managed cloud)
Weaviate	Hybrid (Free if self-hosted; Paid for Weaviate Cloud)	Hybrid (Can run locally in Docker OR managed cloud)
pgvector	Free (Open-source PostgreSQL extension; infrastructure costs apply if hosted)	Hybrid (Local Postgres instance or cloud DB like Supabase/AWS RDS)



Question

If Vector Database Stores the word vectors and ANN is used to search the relevant Vectors, Why we need Embedding models and Embedding Spaces?

Answer: While a Vector Database acts as a storage warehouse for numbers and ANN acts as a fast calculator to measure distances between them, neither actually understands human language. The Embedding Model is the essential translator that converts raw text and user questions into numerical vectors, while the Embedding Space is the mathematical map that arranges those vectors so that concepts with similar meanings sit physically close together. Without the embedding model to translate human thoughts into numbers and the embedding space to give those numbers semantic relationships, a Vector Database would have no meaningful data to store, and ANN would have no logical distances to search.